Standard 12

Time Allowed: 3.00 Hours **CHEMISTRY** Maximum Marks: 70

Part - I

I. Choose the correct answer: 15 x 1 = 15

1. Roasting of sulphide ore gives the gas (A). (A) is a colourless gas. Aqueous solution of (A) is acidic. The gas (A) is
   a) CO_2 b) SO_3 c) SO_2 d) H_2S
2. The basic structural unit of silicate is
   a) $(SiO_3)^{2-}$ b) $(SiO_4)^{2-}$ c) $(SiO)^-$ d) $(SiO_4)^{4-}$
3. Which one among the following allotrope of carbon is used in rock drilling machine?
   a) graphite b) diamond c) fullurenes d) graphene
4. XeF_6 on complete hydrolysis produces
   a) $XeOf_4$ b) XeO_2F_2 c) XeO_3 d) XeO_2
5. Which one of the following interhalogen compound does not exist?
   a) ClF_3 b) FCl_3 c) IF_5 d) BrF_3
6. How many moles of I_2 are liberated when 1 mole of potassium dichromate react with potassium iodide?
   a) 1 b) 2 c) 3 d) 4
7. Silicon carbide is an example of _____ solid.
   a) covalent b) molecular c) ionic d) metallic
8. If 75% of a first order reaction was completed in 60 minutes, 50% of the same reaction under the same conditions would be completed in
   a) 20 minutes b) 30 minutes c) 35 minutes d) 75 minutes
9. Rate = $k[No)^2\ [O_2]^1$ from these rate law the order of the reaction is
   a) 2 b) 3 c) 1 d) 4
10. Dissociation constant of NH_4OH is 1.8×10^{-5} the hydrolysis constant of NH_4Cl would be
    a) 1.8×10^{-19} b) 3.55×10^{-10} c) 5.55×10^{-5} d) 1.80×10^{-5}
11. Assertion : CH_3COOH is disassociated to a small extent
    Reason : Due to common ion effect
    a) Both assertion and reason are true
    b) Assertion is true, reason is false c) Assertion is false, reason is true
    d) Both assertion and reason are false.
12. $HO\text{-}CH_2\text{-}CH_2\text{-}OH$ on heating with the periodic acid gives
    a) methanoic acid b) glyoxal c) methanol d) CO_2
13. Which one of the following is the strongest acid?
    a) 2-nitrophenol b) 4-chlorophenol c) 4-nitrophenol d) 3-nitrophenol
14. Which one among the following ozonolysis reaction product is correctly matched with their reactant?
    a) $CH_2 = CH_2$ - $2\ CH_3CHO$
    b) $CH_3 - CH = CH_2$ - $CH_3CHO + HCHO$
    c) $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}} - CH_3$ - $CH_3COCH_3 + HCHO$
    d) $CH_3 - CH = CH - CH_3$ - $CH_3COCH_3 + CH_3CHO$
15. The reagent is used to distinguish between acetaldehyde and benzaldehyde
    a) Tollens reagent b) Fehlings reagent
    c) 2,4-2 dinitro phenyl hydrazine d) semicarbazide

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Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) 6 x 2 = 12

16. What is mean by Roasting? Give an example.
17. Write the uses of Borax.
18. Write the reason for the anomalous behaviour of nitrogen.
19. Why transition metals producers coordination compounds?
20. Write any three properties of ionic crystals.
21. Write any two examples for zero order reaction.
22. Calculate the pH of 10^{-5} M KOH solution.
23. Write short note about Gatterman Koch reaction.
24. Ethyl magnesium bromide + Ethanol $\xrightarrow{\text{Ether}}$ X $\xrightarrow[H^+]{H_2O}$ Y. Identify X and Y

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) 6 x 3 = 18

25. Write short note about the any one method to purify nickel.
26. Write short note about hydroboration reaction.
27. Why fluorine is more reactive than other halogens?
28. Explain about Frenkel defect.
29. List out the features to affect the rate of the reaction.
30. Explain common ion effect with an example.
31. Write the reaction to convert to the phenol to phenolphthalein.
32. Write the reaction for how cinnamic acid is prepared from benzaldehyde?
33. Give an example reaction for potassium permanganate act as a oxidising agent in the presence of acid medium.

Part - IV

IV. Answer all the questions. 5 x 5 = 25

34. a) Explain the Froth flotation method.

(OR)

b) Complete the following reactions.
i) $B(OH)_3 + NH_3 \rightarrow$ ii) $B_2H_6 + CH_3OH \rightarrow$ iii) $B + NaOH \rightarrow$

35. a) i) Prove chlorine act as the decolourising agent.
ii) Classify and list out the allotropes of sulphur.

(OR)

b) i) $Lu(OH)_3$ and $La(OH)_3$, which is having a high basic nature? Why?
ii) Write the consequences of Lanthanoid contraction.

36. a) i) What is mean by ionic ratio? Give the coordination number of NaCl.
ii) Distinguish tetrahedral and octahedral voids.

(OR)

b) i) Write the Arrhenius equation and explain the terms involved.
ii) The rate of the reaction $X + 2Y \rightarrow$ Product is 4×10^{-3} mol L^{-1} S^{-1}, if [X]=[Y]=0.2 M and rate constant at 400 K is 2×10^{-2} s^{-1}. What is the overall order of the reaction?

37. a) Distinguish between Lewis acid and Lewis base.

(OR)

b) Write about
i) Preparation reaction of red orange dye
ii) Riemer-Tiemann reaction
iii) Dows process

38. a) Write the mechanism of Cannizaro reaction. ***(OR)***
b) Convert the followings:
i) Acetaldehyde → Acetaldol
ii) Acetone → Chloroform
iii) Acetone → Propane

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